

Abiyath Rahman

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PROFESSIONAL SUMMARY

Full-stack software engineer specializing in backend architecture, distributed systems, and cloud deployment. Builds scalable, production-ready applications using Node.js, TypeScript, Redis, MongoDB, and Docker. Experienced designing real-time systems with WebSocket-based synchronization, fault-tolerant job queues, and AI-integrated pipelines, and shipping end-to-end features across the full stack.

EDUCATION

Weber State University

Bachelor of Science in Computer Science.

Ogden, UT

Expected August 2026

EXPERIENCE

Resident Assistant

Aug 2022 – Present

Weber State University

Ogden, UT

- Collaborated with 15 Resident Assistants to improve team response time by 20%, streamlining communication and operations
- Resolved 90% of incidents addressing 10+ personal and academic challenges per semester using conflict resolution strategies
- Managed a community of 200+ students, increasing resident satisfaction by 15% through inclusive programming

PROJECTS

Real-Time Collaborative Code Editor

<https://real-time-collaborative-code-editor-azure.vercel.app>

Node.js, Express.js, MongoDB, Redis, React

- Architected WebSocket-based synchronization system using Socket.io with Operational Transformation for conflict-free text merges across 10+ simultaneous users
- Implemented distributed presence tracking and cursor synchronization using Redis Pub/Sub, enabling horizontal scaling across multiple server instances without session affinity
- Built document persistence layer with MongoDB storing incremental operation logs, supporting full revision history and time-travel debugging
- Integrated Monaco Editor with syntax highlighting for 15+ languages, achieving sub-100ms edit propagation latency under 50 concurrent connections in load testing

Distributed Task Queue System <https://distributed-task-queue-vgzo.vercel.app/>

Node.js, MongoDB, Redis, Docker, React

- Architected distributed job processing system using Redis priority queues and MongoDB-backed state via Mongoose, supporting concurrent worker pools processing 500+ simulated jobs/minute
- Implemented dead-letter collection and exponential backoff retry logic (2s/4s/8s), reducing unhandled job failure rate by 35% against 1,000-job batches
- Built React dashboard with REST API polling at 2-second intervals for job monitoring, queue depth visualization, and system metrics
- Containerized full system with Docker Compose orchestrating API server, 3 worker replicas, Redis, and MongoDB; deployed to Render and Vercel

InsightPulse

Spring Boot, React, PostgreSQL, Docker, AWS Bedrock, Tailwind CSS

- Developed full-stack AI productivity dashboard generating real-time task and calendar insights using Claude v2.1 via AWS Bedrock and Google Calendar API
- Implemented Google OAuth 2.0 with refresh token support and persistent user-specific token storage for seamless reauthentication
- Engineered NLP prompt pipelines and response parsing to summarize calendar workload, detect overload patterns, and surface weekly focus recommendations
- Containerized backend with Docker, deployed to Render; frontend hosted on Vercel with JWT auth and CORS protection

SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL (PostgreSQL, MySQL), C/C++, HTML/CSS, C#

Frameworks: Node.js, Express.js, Socket.io, React, Spring Boot, FastAPI, Flask, Tailwind CSS

Developer Tools: Docker, Redis, Git, AWS (Bedrock), Postman, MongoDB Compass, Render, Vercel, IntelliJ, VS Code

Libraries: Mongoose, ioredis, Socket.io-client, JWT, pandas, NumPy, Matplotlib